

Pratik Raje

+91 9928295768 | rajepratik2407@gmail.com | [LinkedIn](#) | [GitHub](#) | [Leetcode](#) | [Portfolio](#)

WORK EXPERIENCE

Matrice AI (Software Developer-1)

May 2025-Present

- Cut stream startup **latency from ~15s to ~7-9s, replacing a frame-by-frame WebSocket image stream with WebRTC** and optimizing ICE negotiation and media buffering.
- Eliminated a 4-5 second desync between ML overlays and live video by **matching RTP timestamps to async inference within a ~200ms window**.
- Designed a **multi-camera streaming platform** (React/Next.js, WebRTC), capping views at a 4x4 grid so the backend opens 16 ports instead of provisioning 1000+ cameras.
- Engineered a canvas layer (requestVideoFrameCallback, normalized coordinates) rendering 5-25 bounding boxes and polygons per frame at 30 FPS with pan/zoom.
- Developed a **real-time alerting system** (SSE, Fetch streaming) at ~1,000 alerts/day in testing, adding **exponential-backoff** reconnection after restarts left clients silently disconnected.
- Shipped **production frontend applications** in Next.js/TypeScript, developing 12-15 reusable UI components for an analytics platform with role-based access.
- Hardened security with NextAuth (JWT/OAuth), Zod and CSP/HSTS; remediated 6+ CVEs including CRLF-injection and ReDoS, and instrumented Sentry across server and edge runtimes.

PROJECTS

MCU Hub — Next.js, Express, MongoDB

[LIVE](#) · [GitHub](#)

- **Implemented** a graph traversal service over 193 nodes / 755 edges (BFS, weighted Dijkstra, centrality), cutting shortest-path p95 from **34.7ms to 0.2ms** with an in-process adjacency snapshot.
- The win came from **eliminating network round trips, not a better algorithm** — documented with a published benchmark harness and methodology.
- **41 tests** (graph algorithms + route layer against in-memory MongoDB), verified by mutation testing. Route tests caught two production 500s that should have been 400s.
- **Deployed** on Vercel/Render with MongoDB Atlas; CI runs lint, typecheck and tests on Node 20 and 22 per PR.
- **Documented 5 architecture decisions** with rejection criteria, including why not Neo4j — and known limitations rather than omitting them.

SKILLS

- **Frontend:** React, Next.js, JavaScript, TypeScript, HTML5, CSS3, Tailwind CSS, Material UI
- **State Management & Data:** Redux Toolkit, Redux Persist, REST APIs, asynchronous data fetching
- **Real-Time & Video:** WebRTC, WebSockets, Server-Sent Events (SSE), RTP, Canvas API
- **Backend & Databases:** Node.js, Express.js, MongoDB, Mongoose, Zod, Rest API design
- **Developer Tools:** Git, GitHub, Postman, VS Code, Vercel
- **Testing:** Vitest, Supertest, mongodb-memory-server, GitHub Actions (CI/CD)

EDUCATION

Indian Institute of Information Technology, Bhopal

December 2021 - June 2025

Bachelor of Technology in Information Technology (IT)

Relevant Coursework: **DSA, OOPs, Computer Networks, Operating Systems**